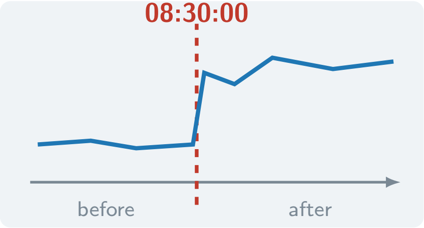
How fast does the market price in the news?

A measurement study. Not a strategy.

`proj-price-discovery` · the model

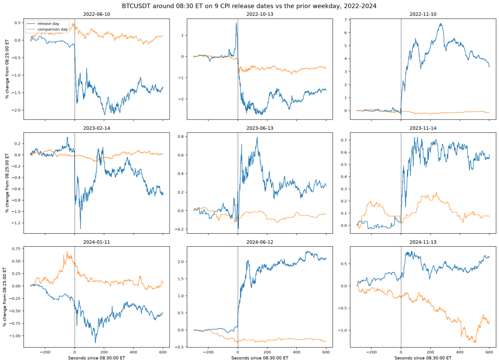
The setup

- CPI lands at **08:30:00 ET**, to the second
- Nobody knows the number one second earlier
- Crypto trades **24/7** — no auction, no open



So the release is a clean shock into a live market.

Does it even react? We checked first.



9 / 9

dates show a jump

5.6×

median vs. a normal day

±

sign follows the surprise

The question, made precise

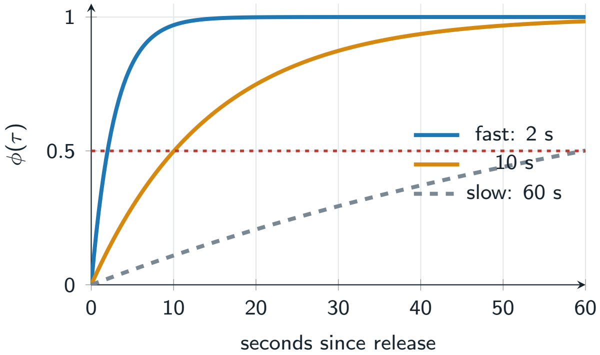
Of the move the release eventually causes,

what fraction has happened by τ seconds?

$$\boxed{\phi(\tau) = \frac{\text{move so far}}{\text{eventual move}}}$$

Report it at 1 s, 10 s, 1 min, 10 min, 1 h. One curve.

What the curve looks like



$$\phi(\tau) = \frac{1 - e^{-\lambda\tau}}{1 - e^{-\lambda H}}$$

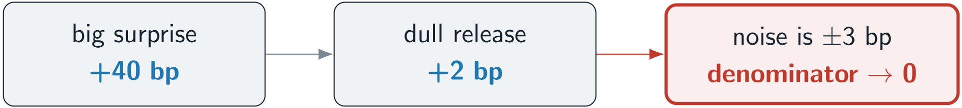
One number per event: λ, reported as the time to get halfway

$$\tau_{1/2} = \frac{1}{\lambda} \ln \frac{2}{1 + e^{-\lambda H}}$$

"half the move happened in 6.9 seconds" — a sentence you can argue with.

The obvious way to compute it

For each event, divide. Then average.



One event can flip the average.

The usual fix is the real trap

"Just drop the events where the price barely moved."

That is selecting on the outcome.

You pick events using information from *after* the release that you never had before it.

The fastest route to a clean-looking wrong answer, and the first thing a reviewer looks for.

So we never divide

Give every event **two** parameters instead of one.

M_e size of the move *don't care*	λ_e rate *the answer*

$$m_e(\tau) = M_e \frac{1 - e^{-\lambda_e \tau}}{1 - e^{-\lambda_e H}} \quad \Longrightarrow \quad \phi_e(\tau) = \frac{1 - e^{-\lambda_e \tau}}{1 - e^{-\lambda_e H}}$$

The size cancels. A dull event now gives a *wide* answer
instead of a divergent one. Nothing is dropped.

212 events. Now what?

No pooling

fit each event alone

212 shrugs

every interval useless

Complete pooling

one λ for all

one number

CPI vs FOMC?

2017 vs 2026?

Partial pooling

own rate,
shared family

both

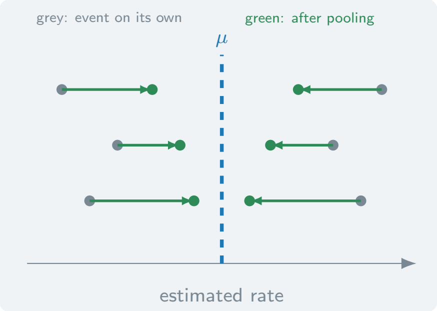
and it self-tunes

$$\log \lambda_e \sim \mathcal{N}(\mu,\ \sigma^2)$$

Own rate per event. All rates from one family.

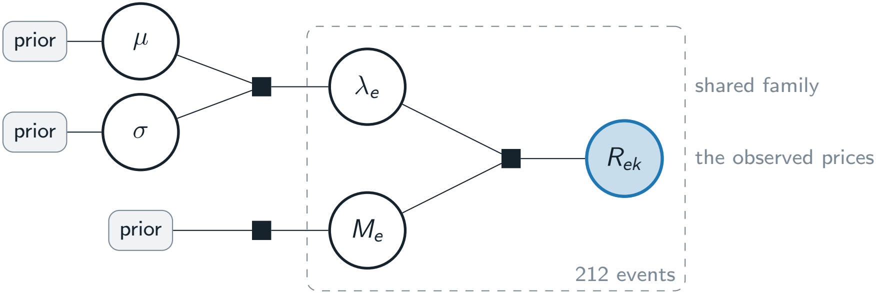
Shrinkage, and why it is not a hack

- Clean event $\rightarrow$ **keeps its own estimate**
- Noisy event $\rightarrow$ **drifts to the crowd**
- The weight is not chosen. It falls out of the two variances.



σ is estimated too. Events differ $\Rightarrow$ σ grows, pooling weakens.
Events are alike $\Rightarrow$ σ shrinks, pooling strengthens. The data decides.

The whole model on one slide



The box repeats per event. Only the shaded node is observed.
The link through μ, σ is how one event's data informs another's estimate.

“Isn’t there just a formula?”

M_e given λ_e plain linear regression	**closed form**
λ_e given M_e λ sits inside $e^{-\lambda\tau}$	**none**
the joint posterior $\sim$650 parameters, no name	**none**

So we sample it.

Hamiltonian Monte Carlo draws from the posterior without ever writing it down.

A 95% interval is literally the 2.5th and 97.5th percentile of the draws.

The output is a table, not an answer

draw	μ	σ	λ_{17}	... 650 cols
1	-2.61	0.83	0.071	
2	-2.48	0.91	0.065	
3	-2.73	0.79	0.088	
$\vdots$	$\vdots$	$\vdots$	$\vdots$	
8000				

Each row is **one complete, self-consistent guess** at every parameter at once.

Population half-time? Transform the μ column, sort, read off the 2.5th and 97.5th percentile. Every reported number is a summary of this table.

First we fit data we made up ourselves

pick a true half-time, say **8 s** → simulate 212 fake events → fit as if it were real → does the interval contain **8 s**?

Repeat 100 times. Honest intervals contain the truth **about 95 times**.

Preregistered criterion: coverage in [90%, 99%].

Why this and not a sanity check. Get the nested-horizon covariance wrong and the intervals come out half as wide as they should be. On real data that looks like an *excellent* result. Here, coverage collapses and the bug is unmissable.

Written down before we look

Fixed in `PREREGISTRATION.md`

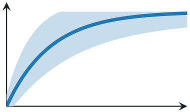
- every prior, with its argument
- the sample and its date range
- which events may be dropped, and why
- all seven robustness checks
- **the result that would falsify us**

What that forecloses

- dropping events that spoil the answer
- widening the horizon until it works
- a robustness check found afterwards, presented as planned
- reporting a null quietly

Merged on its own, before any estimation code exists.
The commit timestamp is what makes the rest of this checkable.

What we will report



the curve

with a 95% band

$\tau_{1/2}$

one number

time to get halfway,

in seconds, with an interval

σ

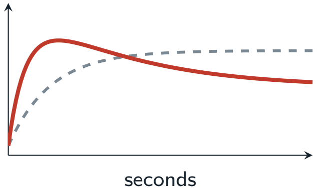
is a finding

large $\sigma \Rightarrow$ there is no

single speed, only a spread

Plus: is CPI faster than FOMC? Has it changed since 2017?
Same model, one extra term.

Where it could break



dashed: what we assume red: overshoot

- **Overshoot.** One rate cannot describe a spike that reverts. **Two panels of slide 2 already look like this.**
- **The hour contains other news.** We cannot separate it.
- **One venue, one coin.** Binance's clock is Binance's.
- **FOMC is out.** Powell speaks at 14:30, inside the window.

Full list, in a hostile reviewer's words: `docs/limitations.md`

The order matters

done
does it react?
9/9 yes

→

done
fix the analysis
preregistered

→

next
validate the model
synthetic data first

Only after recovery and coverage pass do the release-day prices enter the model.